

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
 Second Semester B.Tech Degree Examination June 2022 (2019 scheme)

Course Code: MAT102
Course Name: VECTOR CALCULUS, DIFFERENTIAL EQUATIONS AND TRANSFORMS
(2019-Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

- 1 Find the directional derivatives of $f(x, y) = x^2 - 3xy + y^2$ at the point P(2,1) in the direction of $\vec{a} = \frac{1}{3}\vec{i} + \frac{2}{3}\vec{j}$. (3)
- 2 Evaluate $\int_C 3x y dy$ where C is the line segment joining (0,0) and (1,2). (3)
- 3 Determine the sources and sinks of the vector field $\vec{f}(x, y) = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$. (3)
- 4 Using Divergence theorem evaluate $\iint_S \vec{f} \cdot \vec{n} dS$ where $\vec{f} = 2x\vec{i} + 4y\vec{j} - 3z\vec{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ (3)
- 5 Solve the initial value problem $y'' + 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 2$ (3)
- 6 Solve $y''' - y' = 0$ (3)
- 7 Find the Laplace Transform of $(\sin t + \cos t)^2$ (3)
- 8 Find the inverse Laplace Transform of $\frac{e^{-3s}}{(s+2)^2}$ (3)
- 9 Find the Fourier sine transform of e^{-x} ($x > 0$) (3)
- 10 Find the Fourier Sine Integral of $f(x) = \begin{cases} \sin x & \text{if } 0 < x < \pi \\ 0 & \text{if } x > \pi \end{cases}$ (3)

PART B*Answer one full question from each module, each question carries 14 marks***Module-I**

- 11 a) Find the parametric equation of the tangent line to the curve (7)

$$\vec{r}(t) = 2 \cos \pi t \vec{i} + 2 \sin \pi t \vec{j} + 6t \vec{k} \quad \text{at the point } t = \frac{1}{3}$$

- b) Show that the vector field $\vec{f}(x, y) = 2xy^3 \vec{i} + 3y^2x^2 \vec{j}$ is conservative and find ϕ

such that $\vec{f} = \nabla \phi$. Hence evaluate $\int_{(2,-2)}^{(-2,0)} 2xy^3 dx + 3y^2x^2 dy$. (7)

- 12 a) Find the position and velocity vectors of the particle given $\vec{a}(t) = (t+1)^{-2} \vec{j} - e^{-2t} \vec{k}$, $\vec{v}(0) = 3\vec{i} - \vec{j}$, $\vec{r}(0) = \vec{k}$ (7)

- b) If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and let $\vec{F}(r) = f(r) \vec{r}$ prove that $\text{div} \vec{F} = 3f(r) + \vec{r} \cdot \vec{r}'(r)$ (7)

Module-II

- 13 a) Use Green's theorem to find the work done by the force field

$$\vec{f}(x, y) = xy\vec{i} + \left(\frac{x^2}{2} + xy\right)\vec{j} \quad \text{on a particle that starts at } (4,0) \text{ transverse the upper} \quad (7)$$

semicircle $x^2 + y^2 = 16$ and returns to its starting point along the x-axis.

- b) Find mass of the lamina that is a portion of cone $z = \sqrt{x^2 + y^2}$ that lies between the planes $z = 1$ and $z = 3$, if the density is $\phi(x, y, z) = x^2 z$ (7)

- 14 a) Let σ be the portion of the surface $z = 1 - x^2 - y^2$ that lies above the xy -plane and σ is the oriented upwards. Find the flux of the vector field (7)

$$\vec{F}(x, y, z) = x\vec{i} + y\vec{j} + z\vec{k} \quad \text{across } \sigma.$$

- b) Use Stokes theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y, z) = z^2 \vec{i} + 3x \vec{j} - y^3 \vec{k}$ and C is the circle $x^2 + y^2 = 1$ in the XY plane with counter clockwise orientation looking down the positive Z axis (7)

Module-III

- 15 a) Using the method of undetermined coefficients solve, $y'' - 4y = xe^x$ (7)

- b) Using the Method of variation of parameters solve, $y'' - 4y' + 5y = \frac{e^{2x}}{\sin x}$ (7)

- 16 a) Solve the initial value problem, by method of undetermined coefficients $y'' + 4y = 8x^2$, $y(0) = -3$, $y'(0) = 0$ (7)
- b) Solve the initial value problem $x^2 y'' + 3xy' + y = 0$, $y(1) = -3$, $y'(1) = 1$ (7)

Module-IV

- 17 a) Using Laplace Transform solve $y'' + 5y' + 6y = e^{-t}$ $y(0) = 0$, $y'(0) = 1$ (7)
- b) Using convolution theorem find the Inverse Laplace Transform of $\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$ (7)
- 18 a) Find the inverse Laplace Transform of $\frac{s+8}{s^2 + 4s + 5}$ (7)
- b) Using Laplace Transform solve $y'' + 16y = 4\delta(t - 3\pi)$ $y(0) = 2$, $y'(0) = 0$ (7)

Module-V

- 19 a) Find the Fourier Transform of $f(x) = \begin{cases} e^x & \text{if } -a < x < a \\ 0 & \text{otherwise} \end{cases}$ (7)
- b) Find the Fourier Cosine integral of $f(x) = \begin{cases} \cos x & \text{if } 0 < x < \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$ (7)
- 20 a) Find the Fourier Cosine Transform of $f(x) = \begin{cases} x^2 & \text{if } 0 < x < 1 \\ 0 & \text{if } x > 1 \end{cases}$ (7)
- b) Find the Fourier Transform of $f(x) = \begin{cases} a - |x| & \text{if } |x| < a \\ 0 & \text{otherwise} \end{cases}$ (7)
